Anish Manchanda

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Education

University School of Information Communication & Technology (USICT) — New Delhi, India
B.Tech in Computer Science (Artificial Intelligence), 2024–2028 — *CGPA: 8.75/10*

- **Core CS: Python Programming**, Data Structures, Design & Analysis of Algorithms, Object-Oriented Programming (Java), Operating Systems
- **Systems & AI: Artificial Intelligence & Machine Learning**, **Database Management Systems** (SQL), Computer Networks, Software Engineering (SDLC, Agile, testing)

Bal Bharati Public School — New Delhi, India
Senior Secondary (Class 12): 90.8% | *Secondary (Class 10): 91.6%* — *CBSE Board*

Internships

Pepper Labs AI | *AI/ML Intern* — *June 2026 – August 2026*

- **Integrated multiple LLM providers** (OpenAI, Gemini, Groq, SambaNova) behind a unified abstraction layer with **failure handling and provider fallback**, cutting response latency by **40%**.
- **Diagnosed and fixed a triple-invocation concurrency defect** in the request chain, profiling the call path to cut redundant API calls by **66% (3x → 1x)** and hardening **retry and error-handling** logic.
- **Shipped a production RAG pipeline** to bookbuddy.pepperlabsai.com with hybrid **FAISS + BM25** retrieval, cutting hallucinated responses by **35%**.

MCT India Infotech | *Backend Engineering Intern* — *April 2026 – May 2026*

- **Built backend services for a real-time conversational voice agent** in **Python**, handling session orchestration, concurrent multi-turn token streaming, **REST API** integrations, and data pipelines for live transcripts under call load.
- **Built an autonomous evaluator agent** (**LLM-as-a-Judge**) that assessed live agent conversations against defined quality rules, surfacing *which* interactions failed and *why* to drive automated call-quality monitoring.
- **Designed an automated regression and monitoring framework** running scheduled checks over historical call data to catch behavioral drift before release, cutting manual QA effort by **30%**.

USICT TechSpace | *Computer Vision Intern* — *February 2026 – March 2026*

- **Built internal reporting dashboards** with **Python** data-processing scripts and **Chart.js** visualizations (time-series graphs, distribution pie charts), giving stakeholders visibility into digitization volume and error rates.
- **Developed an OCR data pipeline** in Python digitizing handwritten invoices into a structured **SQL** database feeding those dashboards.

Excel Health-Tech Devices | *Software Engineer Intern* — *December 2025 – January 2026*

- **Built an async scheduler with queue-backed alerting** in **Python/FastAPI** for automated expiry notifications, using WAL mode and thread locks for safe concurrent multi-user writes across a **10+ person team**.

Projects

MaterialMover | *Python, FastAPI, Node.js, React, PostgreSQL, MongoDB, Docker* — Live | GitHub

- **Built an end-to-end full-stack platform**: a **React** frontend over a **Node.js**/Express API and a **FastAPI** service, with JWT auth, unit-tested routes, and **PostgreSQL** and **MongoDB** backing the data layer.
- **Built an agentic AI advisor** using **RAG** with hybrid retrieval over **200+ records**, and deployed the **4-service containerized architecture** via **Docker** with a **CI/CD** pipeline, sustaining **99% uptime** for **50+ users**.

Technical Skills

Languages: **Python**, **TypeScript**/JavaScript, SQL, Java, C/C++, Go
Backend: **FastAPI**, Node.js, Express, REST APIs, microservices, async processing, queueing & retry logic
Voice AI: real-time voice agent backends, **Whisper** speech-to-text, session orchestration, multi-turn streaming, call scheduling, LLM-as-a-Judge evaluation, monitoring
AI & LLMs: LLM integration & provider fallback, **RAG**, prompt engineering, AI agents, FAISS, BM25
Frontend & Dashboards: React, **Chart.js**, data visualization, internal tooling, responsive UI
Data & Infra: **PostgreSQL**, MongoDB, AWS (EC2, S3, Lambda, IAM, RDS), **Docker**, **CI/CD**, Linux
Practices: observability & monitoring, automated testing, code review, documentation, Agile